



(Current) Python Packages for Data Science



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NumPy

- NumPy provides support for large, multi-dimensional arrays and matrices.
- Offers a wide range of mathematical functions to operate on arrays efficiently.
- Key for handling numerical operations in data science workflows.
- Fast and memory-efficient compared to regular Python lists.
- Data science requires efficiency to be useful
- Use NumPy for numerical computations and array manipulations.



Pandas

- Built on NumPy
- Used for handling and analyzing structured data
- Ideal exploring large datasets
- Use Pandas for loading, cleaning, and preparing data



Matplotlib

- Basic plots: line charts, bar charts, scatter plots, histograms, etc.
- Customizable plots: colors, labels, scales, annotations
- Supports publication-quality plots
- Works with Jupyter Notebooks for interactive plotting (people love clicking on things)
- Matplotlib for visualizing insights and trends in the data.



Common packages you'll see around

Name	Common alias	Description
numpy	np	Efficient array computation
pandas	Pd	Data manipulation & analysis
sklearn	sk	Machine learning tools
matplotlib.pyplot	plt	Data visualization & plotting
seaborn	sns	Statistical data visualization
scipi.stats	sp.stats	Statistical tools & probability dists distributions
statsmodels	sm	Statistical modeling & hypothesis testing



Questions?